

Acceptance Testing

UAT Execution & Report Submission

Date	30 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the OptiCrop: Smart Agricultural Production Optimization Engine project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	10	4	2	3	20
Duplicate	1	0	3	0	4
External	2	3	0	1	6
Fixed	11	2	4	20	37
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	5	2	1	8
Totals	24	14	13	26	77

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Data Collection & Preprocessing Module	7	0	0	7
ML Model Building (scikit-learn)	51	0	0	51
Crop Recommendation Engine	2	0	0	2
Input Validation (N,P,K,pH,Humidity)	3	0	0	3

Crop Suitability Assessment	9	0	0	9
Flask Web Application	4	0	0	4
API Integration & Deployment	2	0	0	2

4. Conclusion

The User Acceptance Testing for the OptiCrop: Smart Agricultural Production Optimization Engine has been completed successfully. The system was evaluated across all seven functional modules covering data collection and preprocessing, machine learning model building, crop recommendation, input validation, crop suitability assessment, Flask web application, and API integration. A total of 78 test cases were executed with zero failures and zero untested scenarios, achieving a 100% pass rate.

A total of 77 defects were identified and tracked throughout the testing lifecycle. Of these, 37 defects were fixed before the UAT release, 20 were classified as By Design (accepted system behavior), 8 were marked as Won't Fix with stakeholder agreement, 6 were attributed to external dependencies, 4 were duplicate entries, 1 was not reproduced, and 1 was skipped. No open critical (Severity 1) defects remain unresolved at the time of this report.

The OptiCrop system accurately recommends the most suitable crop based on soil and environmental parameters including Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall. All three key scenarios — Smart Crop Recommendation for Farmers, Crop Suitability and Environmental Assessment, and Agricultural Research and Policy Planning — were validated and confirmed to be functioning as per the defined requirements.

Based on the UAT results, the OptiCrop application meets all defined acceptance criteria and is functionally stable, reliable, and ready for production deployment. The team recommends proceeding with the final release of the OptiCrop Smart Agricultural Production Optimization Engine.